Responsive Web Design

Project: **Hero Archive**
Topic: a superhero fan guide
Date: 4 October 2026

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Group: IT-2511

Repository: https://github.com/bl1rk/WEB2.git
Source folder: assignment3
Live website (open in browser):
https://bl1rk.github.io/WEB2/assignment3/

Project overview
The page has five sections: an introduction, an about section, three strengths, three hero cards, and a reading-guide FAQ. The strengths section uses custom CSS Grid and media queries. The other sections use the Bootstrap grid.

Setup
Bootstrap 5.3.8 CSS is loaded from the official documented CDN URL. The local style.css comes after Bootstrap. The HTML includes the viewport meta tag. The Bootstrap bundle is included before the closing body tag to power the navbar and accordion.

How to run
Open the live website link above from any computer with internet access. The site is hosted on GitHub Pages and does not require the student's computer to be running. For a local copy, open assignment3/index.html. No build command is required.

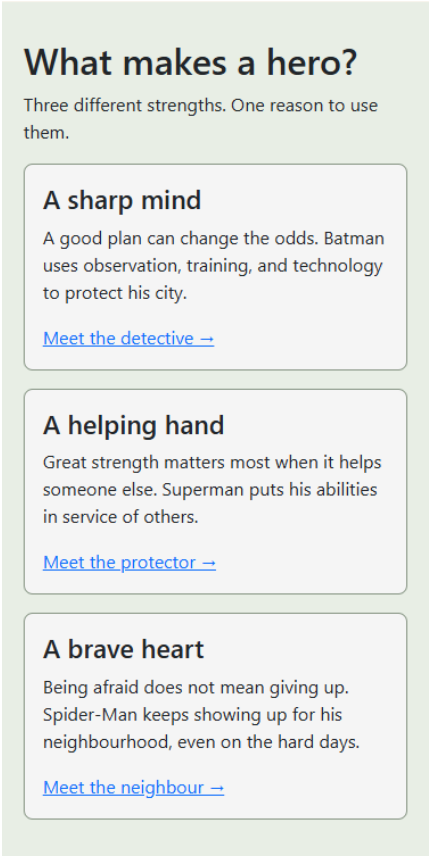
Before submission
The source code is published in the repository linked above. Review the explanations against the code and adapt them into your own words. The screenshots below were captured from the actual local page.

Task 1 / Custom media queries

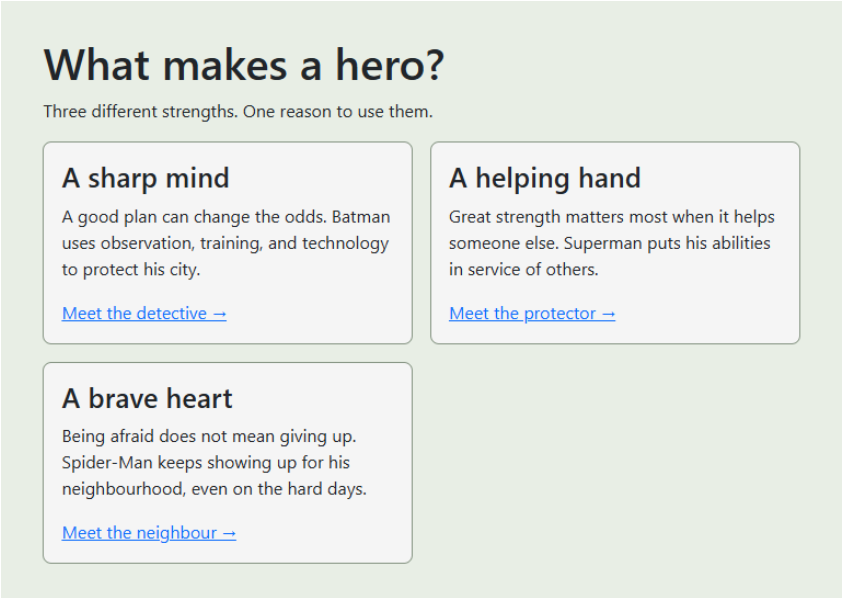
The green **What makes a hero?** section uses only custom classes. Its base grid has one column. At **768 px**, the grid changes to two columns; at **992 px**, it changes to three. The heading also grows from 32 px to 40 px at 768 px, while spacing stays the same.

```
grid-template-columns: 1fr;
@media (min-width: 768px) { /* repeat(2, 1fr) */ }
@media (min-width: 992px) { /* repeat(3, 1fr) */ }
```

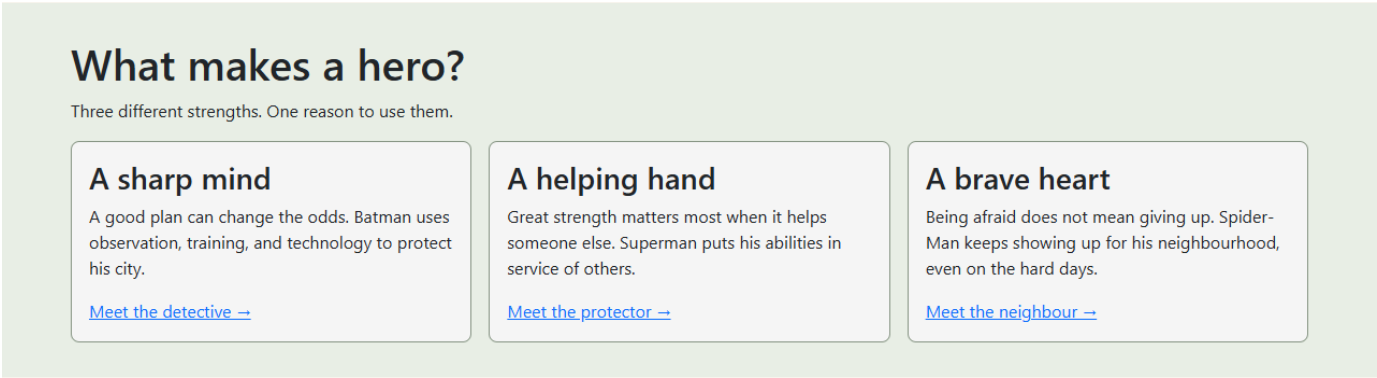
375 px: one column



768 px: two columns



1280 px: three columns



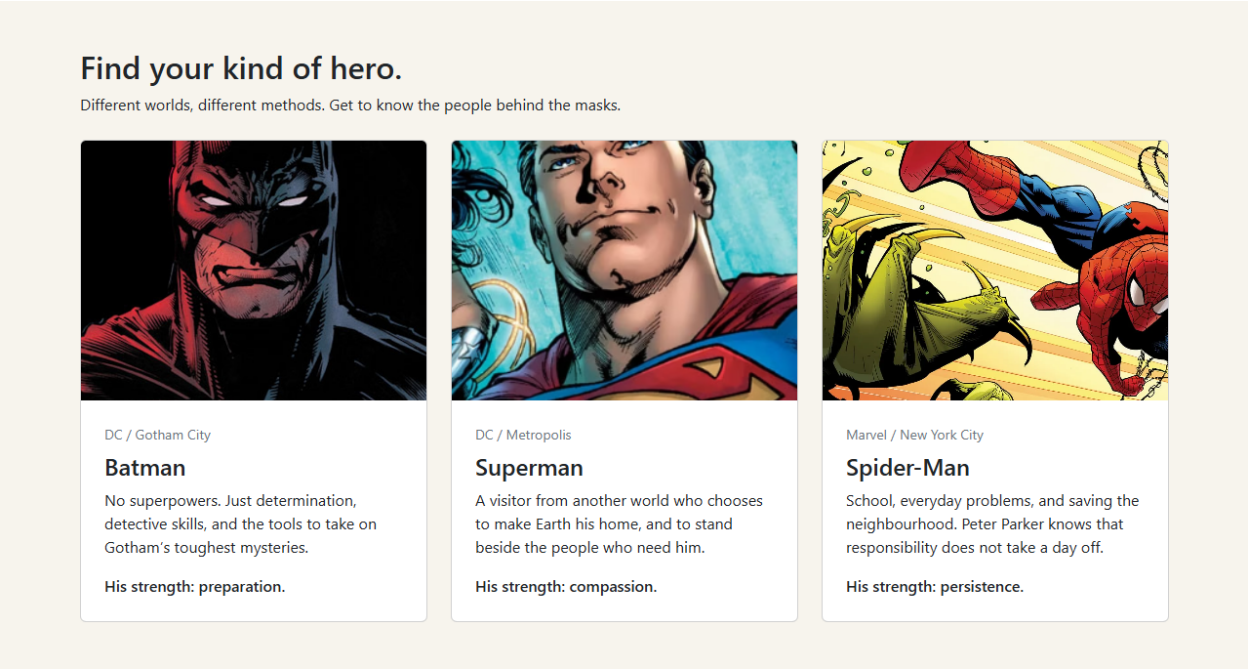
Task 2 / Bootstrap layout

The hero cards use **col-12 col-md-6 col-lg-4**: 12 of 12 columns on a phone, 6 of 12 on a tablet, and 4 of 12 on a laptop. Containers align the content; rows provide the grid; **g-4** adds gutters. Cards use **h-100** and **d-flex flex-column** for consistent heights.

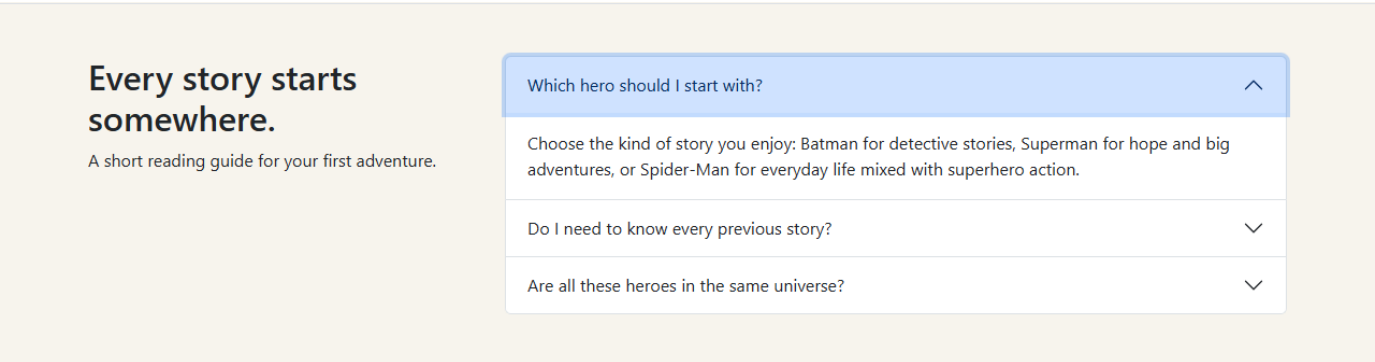
The hero text uses **order-1 order-lg-2**; the image uses the opposite order. On phones, the text comes first and the columns stack. At 992 px they sit side by side with the image first. **d-none d-md-block** hides the longer note on phones, and **d-md-none** shows a shorter version.

The **navbar-expand-lg** navigation collapses below 992 px. Its button targets #mainNav. The reading guide uses the Bootstrap **accordion**; data-bs-parent keeps only one answer open. Buttons were tested by opening and closing the menu and switching answers.

Bootstrap cards at 1280 px



Accordion component at 1280 px



Task 3 / Comparing the approaches

Aspect	Task 1: custom CSS	Task 2: Bootstrap
Layout code	The strengths grid needs a base rule and two media queries.	Hero cards use col-12 col-md-6 col-lg-4 in the HTML.
Design control	The 768 px query changes the column count and heading size.	The card layout follows md/lg breakpoints; custom CSS adds colours and image proportions.
Speed of work	Each grid state must be written and checked manually.	The existing grid, navbar and accordion reduce the layout and interaction code.
Making changes	Changing repeat(3, 1fr) updates the laptop grid in one CSS rule.	Changing col-lg-4 to col-lg-6 makes the cards two per row on laptops.

Custom media queries suit a section with specific layout and spacing changes, such as the strengths section. Bootstrap suits repeated grids and standard interactive components, such as the hero cards, navigation and reading guide. Combining them gives a quick overall layout with control over the distinctive section.

Verification results
At 375, 768 and 1280 px, both grids showed 1, 2 and 3 columns respectively. The document width matched the viewport at all three sizes. Images loaded, section links resolved, the menu and accordion worked, and no JavaScript runtime errors were recorded.

Conclusion
This page demonstrates how mobile-first rules progressively add columns, how the Bootstrap 12-column grid expresses the same layout, and why responsive testing must include navigation, reading order and overflow as well as card widths.

References and assets
Bootstrap 5.3: Introduction; Bootstrap: Grid.
Images were reused from the existing WEB2/images folder. Their original image sources were not supplied. Character artwork belongs to its respective rights holders.

Whole page / 375 px

There's more behind the *cape.*

Extraordinary powers. Very human choices. Explore the characters who remind us what courage looks like.

Explore the heroes →

Your next favourite hero starts here.



In focus: Batman — Gotham's detective

Not every hero can fly.

Some rely on a brilliant mind. Others have powers they never asked for. What connects them is the choice to help when it matters.

Hero Archive is a small fan-made collection. Start with a strength that interests you, meet three memorable characters, and find your own way into their stories.

What makes a hero?

Three different strengths. One reason to use them.

A sharp mind

A good plan can change the odds. Batman uses observation, training, and technology to protect his city.

Meet the detective →

A helping hand

Great strength matters most when it helps someone else. Superman puts his abilities in service of others.

Meet the protector →

A brave heart

Being afraid does not mean giving up. Spider-Man keeps showing up for his neighbourhood, even on the hard days.

Meet the neighbour →

Find your kind of hero.

Different worlds, different methods. Get to know the people behind the masks.



DC / Gotham City

Batman

No superpowers. Just determination, detective skills, and the tools to take on Gotham's toughest mysteries.

His strength: preparation.



DC / Metropolis

Superman

A visitor from another world who chooses to make Earth his home, and to stand beside the people who need him.

His strength: compassion.



Marvel / New York City

Spider-Man

School, everyday problems, and saving the neighbourhood. Peter Parker knows that responsibility does not take a day off.

His strength: persistence.

Every story starts somewhere.

A short reading guide for your first adventure.

- **Which hero should I start with?** — Choose the kind of story you enjoy: Batman for detective stories, Superman for hope and big adventures, or Spider-Man for everyday life mixed with superhero action.
- Do I need to know every previous story?
- Are all these heroes in the same universe?

HERO ARCHIVE.

A student fan project. Characters and artwork belong to their respective owners.

Back to the top ↑

Whole page / 768 px

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A field guide for curious readers and lifelong fans.



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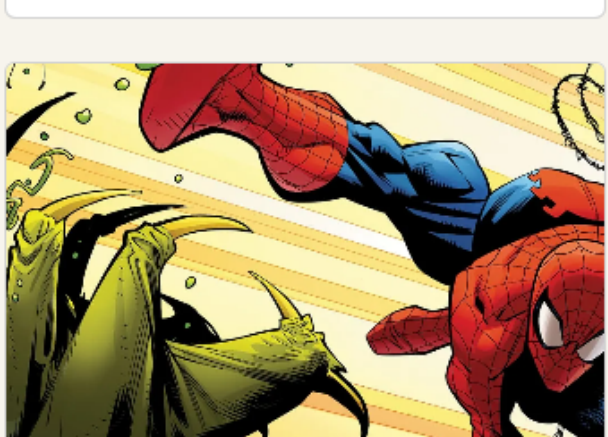


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Whole page / 1280 px

HERO ARCHIVE. The archive · Super strengths · Meet the heroes · Reading guide



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